

EX.NO:1**BASICS OF UNIX COMMANDS****AIM:**

To study and execute Unix commands.

PROCEDURE:

Unix is security conscious, and can be used only by those persons who have an account.

Telnet (Telephone Network) is a Terminal emulator program for TCP/IP networks that enables users to log on to remote servers.

To logon, type telnet server_ipaddress in run window.

User has to authenticate himself by providing username and password. Once verified, a greeting and \$ prompt appears. The shell is now ready to receive commands from the user. Options suffixed with a hyphen (–) and arguments are separated by space.

GENERAL COMMANDS

Command	Function
date	Used to display the current system date and time.
date +%D	Displays date only
date +%T	Displays time only
date +% Y	Displays the year part of date
date +% H	Displays the hour part of time
cal	Calendar of the current month
calyear	Displays calendar for all months of the specified year
calmonth year	Displays calendar for the specified month of the year
who	Login details of all users such as their IP, Terminal No, User name,
who am i	Used to display the login details of the user
tty	Used to display the terminal name
uname	Displays the Operating System
uname –r	Shows version number of the OS (kernel).
uname –n	Displays domain name of the server
echo "txt"	Displays the given text on the screen
echo \$HOME	Displays the user's home directory
bc	Basic calculator. Press Ctrl+d to quit
lpfile	Allows the user to spool a job along with others in a print queue.
man cmdname	Manual for the given command. Press q to exit
history	To display the commands used by the user since log on.
exit	Exit from a process. If shell is the only process then logs out

DIRECTORY COMMANDS

Command	Function
pwd	Path of the present working directory
mkdir	A directory is created in the given name under the current directory
mkdir1 dir2	A number of sub-directories can be created under one stroke
cd subdir	Change Directory. If the subdirstarts with / then path starts from root (absolute) otherwise from current working directory.
Cd	To switch to the home directory.
cd /	To switch to the root directory.
cd..	To move back to the parent directory
rmdirsubdir	Removes an empty sub-directory.

FILE COMMANDS

Command	Function
cat >filename	To create a file with some contents. To end typing press Ctrl+d . The >symbol means redirecting output to a file. (<for input)
cat filename	Displays the file contents.
cat >>filename	Used to append contents to a file
cpsrc des	Copy files to given location. If already exists, it will be overwritten
cp -i src des	Warns the user prior to overwriting the destination file
cp -r src des	Copies the entire directory, all its sub-directories and files.
mv old new	To rename an existing file or directory. -i option can also be used
mv f1 f2 f3 dir	To move a group of files to a directory.
mv -v old new	Display name of each file as it is moved.
Rmfile	Used to delete a file or group of files. -i option can also be used
rm *	To delete all the files in the directory.
rm -r *	Deletes all files and sub-directories
rm -f *	To forcibly remove even write-protected files
Ls	Lists all files and subdirectories (blue colored) in sorted manner.
Lsname	To check whether a file or directory exists.
lsname*	Short-hand notation to list out filenames of a specific pattern.
ls -a	Lists all files including hidden files (files beginning with .)
ls -x dirname	To have specific listing of a directory.
ls -R	Recursive listing of all files in the subdirectories

<code>ls -l</code>	Long listing showing file access rights (read/write/execute- rw x for user/group/others- ugo).
<code>cmpfile1 file2</code>	Used to compare two files. Displays nothing if files are identical.
<code>Wcfile</code>	It produces a statistics of lines (l), words(w), and characters(c).
<code>chmodperm file</code>	Changes permission for the specified file. (r=4, w=2, x=1) chmod 740 file sets all rights for user, read only for groups and no rights for others

OUTPUT

GENERAL COMMANDS

[student@veccse ~]date

Sat May 16 06:10:34 UTC 2020

[student@veccse ~]date +%D

05/16/20

[student@veccse ~]date +%T

10:13:11

[student@veccse ~]date +%Y

2020

[student@veccse ~]date +%H

10

[student@veccse ~]cal

May 2020

Su Mo Tu We Th Fr Sa

1 2

3 4 5 6 7 8 9

10 11 12 13 14 15 16

17 18 19 20 21 22 23

24 25 26 27 28 29 30

31

[student@veccse ~]cal 2020

2020

January

February

March

Su Mo Tu We Th Fr Sa

Su Mo Tu We Th Fr Sa

Su Mo Tu We Th Fr Sa

1 2 3 4

1

5 6 7 8 9 10 11

2 3 4 5 6 7 8

1 2 3 4 5 6 7

12 13 14 15 16 17 18

9 10 11 12 13 14 15

8 9 10 11 12 13 14

19 20 21 22 23 24 25

16 17 18 19 20 21 22

15 16 17 18 19 20 21

26 27 28 29 30 31	23 24 25 26 27 28 29	29 30 31
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April	May	June
Su Mo Tu We Th Fr Sa	Su Mo Tu We Th Fr Sa	Su Mo Tu We Th Fr Sa
1 2 3 4	1 2	1 2 3 4 5 6
5 6 7 8 9 10 11	3 4 5 6 7 8 9	7 8 9 10 11 12 13
12 13 14 15 16 17 18	10 11 12 13 14 15 16	14 15 16 17 18 19 20
19 20 21 22 23 24 25	17 18 19 20 21 22 23	21 22 23 24 25 26 27
26 27 28 29 30	24 25 26 27 28 29 30	28 29 30
	31	

July	August	September
Su Mo Tu We Th Fr Sa	Su Mo Tu We Th Fr Sa	Su Mo Tu We Th Fr Sa
1 2 3 4	1	1 2 3 4 5
5 6 7 8 9 10 11	2 3 4 5 6 7 8	6 7 8 9 10 11 12
12 13 14 15 16 17 18	9 10 11 12 13 14 15	13 14 15 16 17 18 19
19 20 21 22 23 24 25	16 17 18 19 20 21 22	20 21 22 23 24 25 26
26 27 28 29 30 31	23 24 25 26 27 28 29	27 28 29 30
	30 31	

October	November	December
Su Mo Tu We Th Fr Sa	Su Mo Tu We Th Fr Sa	Su Mo Tu We Th Fr Sa
1 2 3	1 2 3 4 5 6 7	1 2 3 4 5
4 5 6 7 8 9 10	8 9 10 11 12 13 14	6 7 8 9 10 11 12
11 12 13 14 15 16 17	15 16 17 18 19 20 21	13 14 15 16 17 18 19
18 19 20 21 22 23 24	22 23 24 25 26 27 28	20 21 22 23 24 25 26
25 26 27 28 29 30 31	29 30	27 28 29 30 31

[student@veccse ~]cal 2020

July 2020

Su Mo Tu We Th Fr Sa
1 2 3 4
5 6 7 8 9 10 11
12 13 14 15 16 17 18
19 20 21 22 23 24 25
26 27 28 29 30 31

[student@veccse ~]who

studentpts/1 May 16 10:05 (172.16.1.14)

[student@veccse ~]who am i

studentpts/1 May 16 10:05 (172.16.1.14)

[student@veccse ~]tty

/dev/pts/1

[student@veccse ~]uname

Linux

[student@veccse ~]echo "hello"

hello

[student@veccse ~]echo \$HOME

/home/student

[student@veccse ~]bc

bc 1.06

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For details type `warranty'.

[student@veccse ~]man lp

lp(1) Easy Software Products lp(1)

NAME

lp - print files cancel - cancel jobs SYNOPSIS

lp [-E] [-c] [-d destination] [-h server] [-m] [-n num- copies] [-o option] [-q priority] [-s] [-t title] [-H handling] [-P page-list] [file(s)]

lp [-E] [-c] [-h server] [-i job-id] [-n num-copies] [-o option] [-q priority] [-t title] [-H handling] [-P page-list] cancel [-a] [-h server] [-u username] [id] [destination] [destination-id]

DESCRIPTION

lp submits files for printing or alters a pending job. Use a filename of "-" to force printing from the standard input.

cancel cancels existing print jobs. The -a option will remove all jobs from the specified destination.

OPTIONS

The following options are recognized by lp:

[student@veccse ~]history

1 date

```
2    date +%D
3    date +%T
4    date +%Y
5    date +%H
6    cal
7    cal 2020
8        cal 7 2020
10   who
11   who am i
12   tty
13   uname
14   uname -r
15   uname -n
16   echo "helloi"
17   echo $HOME
18   bc
19   man lp
20   history
```

DIRECTORY COMMANDS

```
[student@veccse]$ pwd
/home/student
[student@veccse ~]mkdir san
[student@veccse ~]mkdir s1 s2
[student@veccse ~]ls
s1 s2 san
[student@veccse ~]cd s1
[student@veccse s1]$ cd /
[student@veccse /]$ cd . .
[student@veccse /]$ rmdir s1
[student@veccse ~]$ ls
s2 san
```

FILE COMMANDS

```
[student@vecit ~]$ cat>test
```

```
hi welcome operating systems lab
```

```
[student@vecit ~]$ cat test
```

```
hi welcome operating systems lab
```

```
[student@vecit ~]$ cat>>test fourth semester
```

```
[student@vecit ~]$ cat test
```

```
hi welcome operating systems lab fourth semester
```

```
[student@vecit ~]$ cat>test1
```

```
[student@vecit ~]$ cp test test1
```

```
[student@vecit ~]$ cat test1
```

```
hi welcome operating systems lab fourth semester
```

```
[student@vecit ~]$ cp -i test test1 cp: overwrite `test1'? y
```

```
[student@vecit ~]$ cp -r test test1
```

```
[student@vecit ~]$ ls
```

```
s s2 san swap.sh temp.sh test TEST test1
```

```
[student@vecit ~]$ mv san san1
```

```
[student@vecit ~]$ ls
```

```
s s2 san1 swap.sh temp.sh test TEST test1
```

```
[student@vecit ~]$ mv test test1 san1
```

```
[student@vecit ~]$ mv -v san1 sannew
```

```
`san1' -> `sannew'
```

```
[student@vecit ~]$ ls
```

```
s s2 sannew swap.sh temp.sh TEST
```

```
[student@vecit ~]$ cmp test test1
```

```
cmp: test: No such file or directory
```

RESULT

Thus the study and execution of Unix commands has been completed successfully.